Ethical Standards for Information Systems Professionals: A Case for a Unified Code

By: Effy Oz
Department of Accounting
School of Business Administration
Wayne State University
207 Rands House
Detroit, Michigan 48202 U.S.A.

Abstract

Professions have adopted ethical codes and codes of conduct. Physicians, lawyers, and engineers have moral responsibilities and know to whom they are responsible. Professionals in the information systems field need similar guidance. Unfortunately, multiple professional standards in the field are offered by individual organizations. Many of the precepts in these codes are similar, but some are not. This article presents the principles of five codes: three U.S. codes, the Canadian code, and the British code. After an examination of their similarities and differences, it is proposed that we (1) resolve the differences and (2) adopt a single, coherent, international code of ethics for the information systems community.

Introduction

Computer-related legislation started in the late 1970s. However, the need for ethical behavior among computer professionals was already recognized by the late 1960s as the use of computers quickly spread in academic and business organizations. Because computer laws did not exist, professional organizations initiated their own ethical codes.

Some ethical concerns have been resolved in the form of new or amended laws. Some will be addressed by future legislation. Yet many issues will remain to be dealt with by the individual professional. Many professional groups have adopted ethical codes. The emergence of the information systems (IS) professional spurred the major organizations of computer professionals to draft their own codes.

All physicians solemnly swear to heed the Hippocratic oath. All lawyers in the same state, or country, vow to abide by the same ethical standards. However, not all IS professionals are bound by the same set of rules. The reason is simple: the law does not require certification of computer professionals. Certification is voluntary at most. Many IS professionals do not belong to any organization. Membership in a professional organization could, at least, make the member aware of the organization's ethical code. Worse yet, those organizations that have established ethical codes have failed to collaborate and formulate one set of widely accepted rules.

In this article the ethical codes of the five largest IS professional organizations are analyzed and compared. This is done in the hope of inspiring these and other organizations to eliminate differences and establish a "Hippocratic oath" for the entire IS community.

The Organizations

The following is a brief introduction of the professional organizations whose ethical codes are evaluated.

DPMA

The Data Processing Management Association (DPMA) is dedicated to the professional development of the IS professional. Founded in 1951, the

organization now has about 35,000 members throughout the U. S., Canada, and 35 other countries. Its mission is "to advocate effective, responsible management of information to the benefit of its members, employers, and the business community." The organization has adopted a Code of Ethics and Standards of Conduct.

ICCP

The Institute for Certification of Computer Professionals (ICCP) encourages IS professionals to take its exams to ensure proper knowledge and professionalism. It is currently the only U.S. certificate-granting organization in the field. The certificates offered are: Associate Computer Professional (ACP), Certified Computer Programmer (CCP), Certified Systems Professional (CSP), and Certified in Data Processing (CDP). Certificate holders must recertify their skills every three years by either retesting or involvement in approved continuing education courses. Since its establishment in 1973, the organization has certified more than 40,000 professionals in the U.S. and other countries. Candidates for the Institute's certification exams must subscribe to its Code of Ethics.

ICCP's ethical code consists of two parts: the Code of Conduct, and the Code of Good Practice. While the former is mandatory in its nature, and its violation may lead to revocation of a certificate, the latter is recommended behavior, and its violation cannot result in such revocation. The code of ethics includes provisions for revocation of a certificate.

ACM

The Association for Computing Machinery (ACM) was established in 1947 as the society for the computing community. It is the oldest educational and scientific computing society. With more than 82,000 members worldwide, it is also the largest professional organization in the information systems industry. Its goals are to develop information processing as a discipline and to promote responsible use of computers. Its purposes are ''to advance the sciences and arts of information processing, to promote the free interchange of information among specialists and the public, and to develop and maintain the integrity and competence of individuals in the field." Members are expected to adhere to the ACM Code of Ethics and Professional Conduct. An extensively revised Code has been recently introduced to address new problems, such as hacking and computer viruses.

CIPS

The Canadian Information Processing Society (CIPS), founded in 1958, has evolved into Canada's largest association of computer professionals, with more than 6,000 members. The association promotes professionalism among information systems practitioners. It started a certification program in 1989. Currently, 1,200 of its members hold ISP (Information Systems Professional) certificates. Certified members have to meet standards of education and experience.

BCS

The British Computer Society (BCS) was formed in 1957 and became a full member of England's Engineering Council in 1990. It is the professional body of computer practitioners in the United Kingdom and has more than 34,000 members. The Society is concerned with the development of computing and its effective application. It has assumed responsibilities for education and training, for public awareness, and for standards, quality, and professionalism. Its role is to set the professional standards of competence, conduct, and ethical practice for computing in the United Kingdom. It is an authoritative voice to society, government, and industry on all aspects of IS. It influences legislation on data protection, safety, copyright, and product liability; it also provides experts for international standards committees and expert witnesses for courts and tribunals.

The Society has three degrees of membership: associate, member, and fellow. Members at the three levels are accepted according to their qualifications. Higher status is attained if the member has passed the Society's qualifications examinations or has a degree from a BCS-accredited college or university. Thus, BCS is also an accrediting institution.

Evaluation Framework

It has been proposed that professional codes of ethics should be examined along four types of obligations (Johnson, 1985):

1. Obligations to society,
2. Obligations to employer,
3. Obligations to clients, and
4. Obligations to colleagues and professional organizations.

These obligations serve as a framework in the following analysis. However, the fourth type of obligation may actually be divided into three different, but related, obligation sets: (1) obligations to colleagues, (2) obligations to the organization, and (3) obligations to the profession. Therefore, the ethical codes will be compared along these six commitments. Table 1 provides a summary of the codes.

Obligations to Society

The issue

A professional should always consider the welfare of the public when performing his or her job. Information systems have a great impact on the public's security, privacy, and economic interests. Accepted ethical theories require that if a conflict of obligations arises, the net common good should be favored. This means that the good of the public at large should usually stand above the interests of other, smaller constituencies.

Similarities and differences

Although the wording may be different, all the codes address the following points saying that the IS professional should:

- Educate the public about information technology (IT)
- Protect privacy and confidentiality of information
- Avoid misrepresentation of the member's qualifications
- Avoid misrepresentation of IT
- Obey laws
- Avoid taking credit for others' achievements

Additional elements appear in some of the codes but not in others. Other elements are addressed differently in the codes. For example, DPMA divided its code into two parts: a *Standard of Conduct,* which contains the abstract ethical principles the member agrees to uphold, and the *Standards of Conduct,* which specifies the ideal day-to-day behavior of the member. In the *Code of Ethics,* society is not mentioned per se. The member has an obligation to "my country." The member therefore "shall uphold my nation and shall honor the chosen way of life of my fellow citizens.

In an environment of fast, transborder telecommunication and a "global village," it is disturbing to find an obligation, though patriotic, that does not mention humankind as a whole and the welfare of members of other nations. The obligation is clearly to "my country," not to society in the broader sense of the word. But the organization has members in 37 countries. To whom should a DPMA member be obliged if the member is a British national but heads a project for a French company?

The ACM code addresses recent concerns regarding unauthorized access to computers, viruses, and software piracy by specifically proscribing these activities in its code. Similarly, ICCP members promise not to diminish the effectiveness of a system through "acts of commission or omission." The ACM and BCS codes emphasize the obligation to protect intellectual property.

CIPS goes beyond prohibiting misrepresentation of information. Its members are obliged to not withhold information of public concern.

BCS is the only organization that places the public good above the interests of employers and clients. It mandates that its members ensure that the public good is not prejudiced while fulfilling obligations to employer and clients.

Although all five codes require their members to obey laws, this is an area where the codes actually contradict. Under the DPMA code the member must obey local, state, and federal laws (ostensibly those of the U.S.). Under the CIPS code, the member must obey "the law of the country." The ACM code clearly states that "violation of a law may be ethical when that law or rule has an inadequate moral basis or when it conflicts with another law judged to be more important." If you are a member of both DPMA and ACM, whose precept should you obey?

Table 1. Characteristics of Professional Ethical Codes

	DPMA	ICCP*	ACM*	CIPS	BCS*
Obligation to:					
Society	Obligation to "my country"; Protect privacy and confidentiality of information; Inform the public of own expertise; Ensure products of own work are used in socially responsible way; Obey (all) laws; Never misrepresent or withhold information of public concern; Not use confidential knowledge without authorization or for personal gain.	Combat ignorance about IT; Give opinion only in area of expertise; Protect individual privacy; Further public education; Not diminish the effectiveness of a system through acts of commission or omission; Not exaggerate capability of IT.	Contribute to society and human well-being; Avoid harm to others; Be honest and trustworthy; Be fair and take action not to discriminate; Honor property rights including copyrights and patents; Give credit for intellectual property; Access computing and communication resources only when authorized to do so; Respect the privacy of others; Respect users' confidentiality; Know and respect existing laws pertaining to Professional work; Give comprehensive and thorough evaluations of computer systems and their impacts, with special emphasis on risks; Improve public understanding of computing and its consequences.	Further public knowledge; Give opinion only in area of expertise; Not withhold information of public concern; Challenge false information; Not offer misleading information; Not acquire information that is not rightly mine; Obey laws of the country.	Ensure public good is not prejudiced while fulfilling obligations to employer and clients; Respect laws and regulations, especially with regard to fiscal matters, health and safety, and protection of personal data; Ensure my work doesn't interfere with the rights of third parties; Have regard to effect of systems on human rights; Acknowledge and protect intellectual property.
Employer	Keep personal knowledge up-to-date; Share knowledge with others; Present only factual info. to management; Accept responsibility for own work; Not misuse authority; Not take advantage of others' inexperience; Not use others' lack of knowledge for personal gain; Ensure employer is aware of potential conflicts of interest; Present fair, honest, objective viewpoint; Protect employer's interests at all times; Not use employer's resources for personal gain; Not exploit computer system weakness for personal gain.	Not disclose information that can adversely affect third party; Be accountable for my work and my subordinates' work; Protect individual privacy; Disclose limitation of work; Further my education; Not diminish the effectiveness of a system through acts of commission or omission; Be discrete with employer information; Reject a position that involves conflict of interest, unless parties are informed.	Respect confidentiality; Give comprehensive and thorough evaluations of computer systems and their impacts, with special emphasis on risks.	Faithful service; Accept responsibility for my own work; Disclose to others parts of my work they need to know; Reject work I can't perform to management's satisfaction; Guard employer's private info.; Guard employer's proprietary interest; Respect commercial aspects of my obligations to employer.	Complete work within time and budget; Notify employer if work can't be completed within time or budget so corrective action can be taken; Not disclose confidential information; Exercise competence to the extent I claim; Declare level of competence when my service is sought.

Clients	None	Not disclose information that can adversely affect third party; Protect individual privacy; Disclose limitation of work performed; Not diminish the effectiveness of a system through acts of commission or omission; Not exaggerate capability of IT; Avoid professional jargon; Be discrete with client information; Reject a position that involves conflict of interest, unless parties are informed.	Respect confidentiality; Give comprehensive and thorough evaluations of computer systems and their impacts, with special emphasis on risks; Honor contracts, agreements, and assigned responsibilities.	Frank counsel; Guard confidential information; Protect user of my product or service against loss or harm; Avail expertise; Avoid conflict of interest and give notice of potential ones; Apply obligation to employer to client when I'm a consultant.	Same as obligations to employer; As consultant: notify client in writing when my interest may prejudice theirs; provide a written contract before accepting an assignment; notify client of potential time or budget overruns; and do not invite an employee of the client to consider alternative employment without client consent.
Colleagues	(Not a constituency)	Not denigrate colleagues; Contribute to colleagues' education.	(Not a constituency)	Treat colleagues with respect, hold their right to success as important as mine; Contribute to professional knowledge.	(Not a specific constituency)
The professional organization or its members	Honesty; Take action against others' unethical conduct; Share own knowledge; Cooperate for understanding and problem solving; Not take undue credit.	Report violations of the Code; testify in ethical proceedings; and serve on panels to judge.	Uphold and promote the Code; Agree to take action to remedy if the Code is violated.	(Not a specific constituency)	Avoid acts detrimental to the profession; Apply high professional standards to my social life; Strive to enhance public confidence in the profession; Act with integrity; Not do anything that may adversely affect another member; Not misrepresent the organization; Declare my position when in conflict of interest with the organization.
Profession	Same obligations as to organization members.	Apply high professional standards to my personal life; Exercise competence to the level I claim; Not withhold professional knowledge, unless disclosure harms a third party.	Strive to achieve the highest quality in the process and products of professional work; Acquire and maintain professional competence; Accept and provide appropriate professional review.	Maintain competence; Maintain high personal standards; Give due credit to others	As sole U.K. organization, all that applies to the *Organization* also applies to the *Profession.*
Sanctions	None	Revocation of certificate; Clear procedure.	Termination of membership; Procedure not specified.	Warning, suspension, or revocation of membership; Clear procedure.	Reprimand, suspension, or revocation of membership; Clear procedure.
Guidance to Members	Not offered	Not offered	Not offered	Not offered	Offered
Priority of Constituents	No	No	No	No	No

* The code is not arranged by constituencies but by type or responsibility (e.g., accountability, protection of privacy, etc.). Wherever no constituent is specified in the code, placement of the obligations in the different categories (society, employer, etc.) was determined by applicability.

References: The draft of the ACM Code appears in *Communications of the* ACM (1992). *MIS Quarterly* readers can examine the appendices of Couger (1989) for a full reprint of the ACM and DPMA codes as they existed at that time. The other codes can be obtained directly from the respective organizations. The codes are presented in their entirety in Oz (forthcoming, 1993).

Obligations to Employer

The issue

An employee is paid and trusted by the employer to perform assigned tasks to the best of the employee's ability. Protecting the employer's interests is what people usually mean by "work ethics." If the employee is a professional, the employer's trust is greater because the employee performs activities that require expertise that the employer may not possess. Thus, the employer may not be able to scrutinize the professional.

Similarities and differences

All five codes share a core of obligations to the employer, which include:

- Update my knowledge in the field of IT
- Accept responsibility for my own work
- Present work-related information to the employer in an objective manner
- Respect confidentiality

Only some of the codes include the following points:

- Protect employer's interests
- Inform the employer of, or reject a position that involves, conflicts of interest
- Reject work I cannot perform
- Do not use employer's resources for my own use

The codes differ in that they do not all have the same elements, but they do not contradict each other. All of the shared rules should appear in the unified code because they reflect important protection of the employer against misconduct.

There is one unclear point in the DPMA code. It starts with the words ''I acknowledge that I have an obligation to management." A reasonable person would understand that management is one's employer. Nevertheless, the code goes on to state an obligation to the employer: "I have an obligation to my employer whose trust I hold, therefore, I shall endeavor to discharge this obligation to the best of my ability, to guard my employer's interests, and to advise him or her wisely and honestly." The standards of conduct toward management and standards of conduct toward employer are practically the same and should not be split.

Obligations to Clients

The issue

The business depends on clients for its survival. Failure of an employee to satisfy the employer's contractual and ethical obligations to the client hurts the employer. When the professional serves a client as a consultant, the client's relationship with the professional is similar to that of an employer, with all the ethical implications of such a relationship.

Similarities and differences

DPMA's code ignores the client. ICCP combines employer and client in the same ethical obligations. It states that "certified computer professionals have an obligation to serve the interests of their employers and clients loyally, diligently and honestly." Indeed, employers and clients are similar in the sense that both receive services from the professional. However, it is unrealistic to expect an employee to be equally loyal to a client and the employer. After all, it is the employer who pays for the employee's work. Also, the employee may not have the complete picture of the employer's contractual obligations to the client. And he or she is not directly bound by any contract with the client. For these reasons we cannot expect the employed professional to be as obligated to the client as to the employer.

The situation is different when the professional serves as an independent consultant. In this capacity, the professional assumes full responsibility to perform the task using his or her own opinion as to methods and means. He or she is then directly responsible to the client and bears all the obligations without an intermediary entity.

The BCS code includes a special section specifying the member's ethical obligation when serving as a consultant. ICCP requires the consulting professional to apply the obligations to employer to the client.

The four codes (all but DPMA's) share these obligations to the client:

- Protect confidential information and privacy
- Give comprehensive opinion regarding information systems

At least one, but not all, of the codes includes:

- Avoid, or notify the client of, conflict of interests
- Do not diminish the effectiveness of a system through commission or omission
- Honor contractual obligations
- Avail expertise
- As a consultant, do not employ an employee of the client without the client's consent
- Avoid professional jargon

All of these obligations should be included in the unified code. Except for the second edict, these are obligations of any professional to a client. IS professionals are also required to not abuse their expertise to diminish the effectiveness of their creations. Perhaps positive language would be more appropriate: "Use my expertise to provide the client with the best system within contractual limits."

Professionals have developed their own jargon. The use of such vernacular sometimes allows members of the same trade to more efficiently exchange ideas. However, when a professional uses jargon when communicating with a lay person, the results may be undesirable. Lay people are often reluctant to admit they do not understand jargon terms. While the professional means one thing, the other party may understand another thing. In the IS industry, where systems development is frequently expensive, clear personal communication is of utmost importance. The ICCP took notice. Its code encourages the certified professional to use plain language in communication with clients.

Obligations to clients across the codes do not contradict each other. They are either missing in some codes or emphasized to different degrees.

Obligations to Colleagues

The issue

Members of the same trade share many interests. Thus, one is expected to help one's colleagues and respect their work.

Similarities and differences

In some professions obligations to colleagues are sometimes raised above other obligations. For example, in some countries it is virtually impossible to convince a physician to testify against a fellow doctor in malpractice cases. None of the four codes even tries to foster such "loyalties" among IS professionals. On the contrary, two of the codes encourage members of the organizations to expose unethical acts of their colleagues. This implies that loyalty to the organization, or to the profession in general, is more important than loyalty to a colleague.

A DPMA member has an obligation to fellow members but not to other colleagues. Likewise, the ACM and BCS codes do not mention colleagues as the subject of any ethical obligation of the member.

ICCP and CIPS require the member to respect colleagues and contribute to their professional knowledge.

Obligations to the Organization

The issue

Professional organizations require their members to uphold the organizations' objectives and serve their interests for the common good of all members.

Similarities and differences

Under the DPMA code the member has obligations to fellow DPMA members but not to the organization per se. In the CIPS code, the organization is not addressed as a specific party to which the member has obligations. This may stem from the fact that CIPS regards itself as the sole Canadian IS professional association. Similarly, the BCS does not clearly differentiate between "organization" and "profession," as it is the sole IS professional body in the United Kingdom. But the BCS does include obligations to the organization. It forbids the member to misrepresent the organization. The member must also declare when his or her personal position is in conflict with that of the organization.

ICCP and ACM require that the member report violations of the code. In addition, the ICCP

member agrees to testify in ethical proceedings and sit on judging panels.

Loyalty to an organization is a fine principle but should not exceed obligations to other constituencies. Also, the member should not be subject to edicts that may contradict those of other IS organizations of which he or she may be a member.

Obligations to the Profession

The issue

Ethical obligations to the profession stem from the same reasons for obligations to colleagues. However, a member can help the profession or hurt it, regardless of behavior toward colleagues. For example, repetitive violation of contractual responsibilities may blemish not just the offender but the entire profession. Usually, obligations to the profession as a whole are placed above obligations to a colleague. For example, the member is expected to report an unethical act of a colleague. This follows the notion of preferring the "common good."

Similarities and differences

All five organizations encourage commitment to the data processing profession. DPMA's code does not distinguish between the profession in general and fellow members, and, therefore, the obligations are limited to relationships between colleagues. Its code states the same obligations to "fellow members and the profession."

The other four codes share one responsibility: the application of high professional competence. At least one code, but not all, includes the following elements:

- Apply high professional standards to personal or social life
- Avoid acts detrimental to the profession
- Enhance public confidence in the profession

As the BCS is a sole national organization, its members' obligations to the organization apply to the profession. However, some of the obligations are specifically directed at "the profession." The BCS code is significantly more detailed on this issue than the other codes.

It is recommended that all the above precepts be included in the unified code because all of them contribute to the fostering of high regard for the profession.

Sanctions Against Violations

What happens when a member violates the organization's code of ethics? All of the organizations except DPMA include sanctions against violators. ACM is somewhat vague regarding its policy against violators. Violations are "dealt with in accordance with ACM Policies and Procedures," but these policies and procedures do not mention any measures against violations. The provisions of ICCP and CIPS are clear and detailed. They specify how a complaint should be submitted, the formation of a hearing panel, the hearing process, and the appealing process.

Conflicts, Priorities, and Guidance

One flaw is apparent in all five ethical codes: the lack of priorities among the subjects of moral obligations. Obligation to one party may collide with obligation to another party. Protecting one's employer's interests may harm the public; protecting a colleague's interests may contradict obligations to one's employer; etc. What is the member to do in such cases?

For example, consider this scenario. A programmer working for a consulting firm is involved in a large project for a client. The programmer comes to realize that some of the code she is developing will not be compatible with other systems employed by the client. When she brings the new facts to the employer, he demands that she follow his instructions. Her responsibility to the employer is to obey the instructions. However, her obligation to the client is to inform him about the incompatibility. The programmer is faced with an ethical dilemma to which there is no solution in any of the above ethical codes.

The codes require that the member not divulge confidential information of the client. Now, let's assume that in the course of the member's work, the member received information of acts committed by the client, or the employer, that could harm the public. Should the member abide by the rule?

Should he or she be "more ethical" toward the public, or toward the employer or client?

While ethical conflicts are just as frequent in other professions, the ethical codes of other professions provide clearer decision rules. For example, a lawyer's first obligation is always to his or her client. The same principle applies to physicians. Journalists strictly obey the rule of not exposing a source without the source's consent. Journalists have protected their sources to the point of obstructing justice; namely, they have preferred the source's interests to the public's interests. An architect who learns that a building may not meet safety standards is expected to not proceed with his or her work in order to protect the public even though this may conflict with his or her obligations to the employer or client. Unfortunately, none of the codes discussed here provides any order of priorities. The leaders of the IS professional organizations should try to outline priorities of moral obligations.

A conflict should not always be resolved in favor of the same constituent. Each case should be carefully examined by the individual before making a decision, and the responsibility of judgment should rest with the individual. Still, a code of ethics should try to help make the judgment.

What should the member do when in a quandary? BCS offers guidance. Its code starts with the following words: "The Society, through its Professional Advisory Committee, is ready at all times to give guidance on the application of the Code of Conduct, and any member needing clarification or amplification of his or her obligation for the proper observance of professional conduct should seek the Committee's assistance." The codes of the other organizations do not mention any proposed assistance.

Need for a Unified Code of Ethics

The reason for multiple ethical codes stems from the fact that IS is, relatively, a new field. The professional associations felt they had to develop some ethical standards for the emerging trade. They should be commended for that. There are only slight differences among the objectives of the organizations. All seek the public's respect and appreciation of the special knowledge their members possess. This alone warrants a unified code of ethics.

Perhaps the best way to shape codes into a comprehensive, single code would be by examining the points of similarity between the profession and other professions that have widely accepted coherent ethical standards. Mylott (1986) equates computer professionals to other professions:

> In the services they perform, computer professionals most resemble a combination of accountants, architects, and engineers. Like architects and engineers, computer professionals create specifications and supervise the implementation of specifications. Yet, while architects and engineers rarely construct the buildings they have designed, computer professionals usually create the object of their specifications; they write computer software and propose combinations of hardware and software to purchasers. In order to develop computer software and to assemble configurations of computer hardware and software, computer professionals, like accountants, often perform financial and business analysis (p. 269).

This observation describes us well. Members of the IS community usually regard themselves as a profession. They meet frequently in professional conferences, read and contribute to professional publications, and share a professional jargon.

The relationship between IS professionals and the public is similar to most other professional relationships in terms of knowledge and reliance. People approach a physician for help because the physician has knowledge that they do not have. People hire a lawyer because the lawyer possesses skills that they do not possess. Lay people trust their interests to experts. This is the main reason why a profession needs an ethical code of conduct. Similarly, IS professionals possess expertise that others do not. Their obligations to the public and clients are therefore similar to those of other experts.

Ethical codes from three different countries have been presented here. The similarities among the codes outnumber the differences. As is evident, the differences cannot be attributed to organizational or national interests. IS professionals in one

country do not differ from their colleagues in other countries, and they are all expected to abide by the same standards of practice, regardless of organizational or national affiliation. This is especially true in the reality of a "global village," where computer networks diminish the importance of national borders.

A single, coherent code of professional conduct will better achieve the five objectives that professional codes of ethics are supposed to achieve (Johnson and Snapper, 1985):

- **Inspiration:** to inspire members of the profession to act more ethically
- **Sensitivity:** to encourage the members to be sensitive to the moral aspects of their jobs
- **Discipline:** to enforce certain rules of the profession on its members to achieve integrity
- **Advice:** to provide advice in cases of moral complexity and ethical dilemma
- **Awareness:** to alert employers and clients as to what they can expect of the member when performing his or her job

A single international code would also help in fostering a stronger public recognition of the IT profession and greater commitment on the part of IT professionals toward society, employers, clients, and colleagues.

Recommendations

The International Code of Ethics and Professional Conduct for IS Professionals should consist of the following principles:

1. The code should be organized along constituencies to which the IS professional has obligations, i.e., the public, the employer, the client, the profession, and colleagues. (The codes of ICCP, ACM, and BCS are not arranged by constituencies.) Focus on the constituencies induces a better sense of responsibility than does focus on the areas of responsibility.
2. The obligations to the different constituencies should be a union of all of the obligations as detailed in the above five codes. As the BCS code appears to be the most comprehensive, it may serve as a good basis for the union. Namely, the drafters will add to the BCS code ingredients that it lacks from the other codes.
3. As the unified code is international, it should be free of obligations to any specific country. The professional's obligation to the public is an obligation to humankind, not to the citizens of a country. The countries' laws have to be obeyed by IS professionals anyway, just as any other individuals must. "Colleagues" will mean all colleagues, regardless of nationality.
4. Due to the interorganizational nature of the code, obligations to professional organizations will be identical to obligations to the profession. Therefore, there will be no need for obligations to organizations. If at all necessary, individual organizations will still be able to maintain codes specific to their memberships. These codes should not contradict those of the unified code. (Such codes may relate to the unified code just as state laws relate to the U.S. constitution.)
5. The code should provide general guidelines for priorities of obligations with respect to constituencies. That is, they will state the interests of which constituency the professional should prefer in case of a conflict. For example, the code will say that the welfare of the public at large should always come before the interests of clients, employers, and colleagues. Clearly, IS professionals may find themselves in conflicts for which there are no prescribed solutions, but the guidelines may help in many cases. To resolve such conflicts, organizations should jointly, or separately, establish "ethics teams" that the individual professional can contact for advice. It should be understood by all that this is not legal, but moral, advice. Such advice should be extended only in the absence of applicable law. (For example, only two U.S. states require that a person who knows of a violation of the state's computer crime statute must report the violation. In other states, an IS professional may be reluctant to report a violation because reporting may harm another party more than the party that is harmed would benefit from the reporting.)
6. The code should detail the procedures for reporting and processing complaints against

violators of the code and the measures taken against violators. An ethics code without sanctions is like a crocodile without teeth. Sanctions may include expulsion from professional organizations, revocation of professional certificates, denial of admittance to professional gatherings, and other temporary or life-time penalties.

Conclusion

The organizations discussed in this article are among the world's most prominent IS professional associations. Since they, and other similar organizations, promote the same objectives, there is reason to embark on an effort to form an international code of ethics for IS professionals. The unified code will enhance the public's perception of IS specialists as representing a true, responsible profession. It will also assure the public of the profession's concern for ethical development and implementation of information systems.

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